

Quantum Mechanics & Modern Physics

1. Introduction

Quantum mechanics and modern physics study matter and radiation at atomic, subatomic, and relativistic scales. These topics emerged because classical physics could not explain blackbody radiation, photoelectric effect, atomic stability, discrete spectra, and microscopic behavior.

For BSc Physics, this subject is central because it gives the mathematical and conceptual framework for atoms, nuclei, photons, solids, semiconductors, lasers, and advanced physics.

2. Historical background

Modern physics began with key experiments and ideas that overturned classical assumptions. Planck introduced energy quantization, Einstein explained the photoelectric effect, Rutherford revealed the nuclear atom, Bohr proposed quantized orbits, and de Broglie connected waves and particles.

These developments led to quantum mechanics, which describes microscopic systems probabilistically rather than deterministically.

3. Blackbody radiation

Blackbody radiation could not be explained by classical theory and led to Planck's quantum hypothesis. Energy is emitted or absorbed in discrete packets called quanta.

Planck's relation

$$E = h\nu$$

where h is Planck's constant and ν is frequency.

Significance

This idea marks the beginning of quantum theory and explains the observed spectrum of blackbody radiation.

4. Photoelectric effect

The photoelectric effect shows that light behaves as particles called photons. Electrons are emitted from a metal surface only if the incident light has frequency above a threshold frequency.

Einstein's equation

$$h\nu = \phi + K_{\max}$$

where ϕ is the work function and $K_{\max}$ is maximum kinetic energy of the emitted electron.

Key observations

- No emission below threshold frequency.
- Intensity affects number of photoelectrons, not their maximum kinetic energy.
- Kinetic energy depends on frequency, not intensity.

5. Compton effect

The Compton effect demonstrates particle-like momentum of photons. When X-rays scatter from electrons, the wavelength changes due to energy and momentum conservation.

Compton shift

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta)$$

This provides strong evidence for the photon concept.

6. Wave-particle duality

Wave-particle duality states that microscopic entities such as electrons and photons exhibit both wave-like and particle-like behavior depending on the experiment.

de Broglie hypothesis

$$\lambda = \frac{h}{p}$$

where p is momentum.

Importance

This relation explains electron diffraction and the basis of matter waves.

7. Bohr atomic model

Bohr's model explained the hydrogen spectrum by postulating quantized orbits and angular momentum. Though limited, it is a key stepping stone in modern physics.

Bohr postulates

- Electrons move in stable orbits without radiating energy.
- Only certain orbits are allowed.
- Angular momentum is quantized:

$$mvr = \frac{nh}{2\pi}$$

- Radiation occurs only during transitions between allowed orbits.

Hydrogen spectrum

The spectral lines of hydrogen are explained by energy level differences.

8. Atomic spectra

Atomic spectra arise because electrons occupy discrete energy levels. When electrons transition between levels, photons of specific wavelength are emitted or absorbed.

Series in hydrogen

- Lyman series.
- Balmer series.
- Paschen series.
- Brackett series.
- Pfund series.

9. Uncertainty principle

Heisenberg's uncertainty principle states that certain pairs of observables cannot be measured simultaneously with arbitrary precision.

Position and momentum

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

Energy and time

$$\Delta E \Delta t \geq \frac{\hbar}{2}$$

This principle is fundamental to quantum behavior and rules out classical orbit pictures for electrons.

10. Schrödinger equation

The Schrödinger equation is the central equation of non-relativistic quantum mechanics. It determines the wave function of a system.

Time-dependent form

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi$$

Time-independent form

$$\hat{H}\psi = E\psi$$

Meaning

The wave function ψ is not directly observable, but $|\psi|^2$ gives the probability density.

11. Interpretation of wave function

The wave function contains complete information about the quantum state of the system. It must be single-valued, finite, continuous, and normalizable.

Probability density

$$P(x, t) = |\psi(x, t)|^2$$

This gives the likelihood of finding a particle in a given region.

12. One-dimensional applications

A typical B.Sc. treatment includes the particle in a box, finite potential well, and tunneling concepts.

Particle in a box

The particle is confined to a region with infinitely high walls, leading to quantized energy levels.

Energy levels

$$E_n = \frac{n^2 h^2}{8mL^2}$$

where $n = 1, 2, 3, \dots$ physics.

Tunneling

Quantum tunneling allows particles to pass through barriers that would be forbidden classically. It is important in nuclear decay and semiconductor devices.

13. Operators and observables

In quantum mechanics, physical quantities are represented by operators acting on wave functions. Measurement outcomes correspond to eigenvalues of these operators.

Common operators

- Position operator.
- Momentum operator.
- Hamiltonian operator.
- Angular momentum operator.

14. Angular momentum

Angular momentum is quantized in quantum mechanics. It includes orbital and spin angular momentum.

Quantum numbers

- Principal quantum number n .
- Azimuthal quantum number l .
- Magnetic quantum number m_l .
- Spin quantum number m_s .

Magnitude of orbital angular momentum

$$L = \sqrt{l(l+1)}\hbar$$

15. Spin and Pauli principle

Electron spin is an intrinsic form of angular momentum. The Pauli exclusion principle states that no two electrons in an atom can have the same set of four quantum numbers.

This principle explains electronic structure, chemical periodicity, and atomic stability.

16. Stern-Gerlach experiment

The Stern-Gerlach experiment demonstrated space quantization and supported the concept of spin. It showed that angular momentum components are quantized.

17. Atomic and molecular structure

Modern physics also includes atomic models, X-ray spectra, and molecular bonding ideas that emerge from quantum principles.

18. Nuclear physics

Nuclear physics studies the nucleus, radioactivity, nuclear reactions, binding energy, and nuclear models.

Radioactivity

Radioactive decay includes alpha, beta, and gamma processes. Decay is random but follows exponential laws.

Decay law

$$N = N_0 e^{-\lambda t}$$

and half-life:

$$T_{1/2} = \frac{\ln 2}{\lambda}$$

Binding energy

Binding energy explains nuclear stability and mass defect.

19. Nuclear models

Important models include the liquid drop model and shell model. These explain collective behavior and magic numbers, respectively.

20. Particle physics overview

Modern physics often introduces elementary particles, conservation laws, and interactions. This helps students understand the larger structure of matter.

21. Special relativity

Special relativity reshaped modern physics by introducing time dilation, length contraction, and mass-energy equivalence.

Key relation

$$E = mc^2$$

This expresses the equivalence of mass and energy.

22. Applications of modern physics

Modern physics underlies many technologies:

- Lasers.
- Semiconductors.
- Nuclear reactors.
- Medical imaging.
- Atomic clocks.
- Quantum devices.
- Spectroscopy.
- Electron microscopy.

23. Key formulas

- $E = h\nu$.
- $h\nu = \phi + K_{\max}$.
- $\Delta\lambda = \frac{h}{m_e c} (1 - \cos \theta)$.
- $\lambda = \frac{h}{p}$.
- $\Delta x \Delta p \geq \frac{\hbar}{2}$.
- $i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi$.
- $E_n = \frac{n^2 h^2}{8mL^2}$.
- $N = N_0 e^{-\lambda t}$.
- $T_{1/2} = \frac{\ln 2}{\lambda}$.
- $E = mc^2$.

24. Exam-oriented distinctions

Classical vs quantum

- Classical physics is deterministic.
- Quantum physics is probabilistic.

Particle vs wave

- Particles are localized with momentum.
- Waves show interference and diffraction.
- Quantum objects show both depending on measurement.

Orbit vs orbital

- Orbit is a classical path.
- Orbital is a probability distribution.

25. Summary

Quantum mechanics and modern physics form the conceptual core of modern BSc Physics. They explain why classical ideas fail at microscopic scales and provide the mathematical framework for atoms, nuclei, semiconductors, radiation, and many advanced technologies.